

The Treasury Risk Triangle: A Public-Data Decomposition of Fixed-Income Exposures

Anonymous

June 24, 2026

Abstract

We decompose daily U.S. Treasury market return variation across 14 liquid fixed-income instruments into three economically distinct, publicly observable risk components: the model-implied real-rate change (R), the model-fitted inflation-compensation change (BEI), and the residualized nominal-curve shape (Shape). R and BEI are taken from the D’Amico–Kim–Wei (DKW) TIPS decomposition; Shape is constructed from the Gürkaynak–Sack–Wright (GSW) nominal yield curve. Both data sources are public Federal Reserve releases. We estimate per-asset HAC Newey–West linear projections at the daily horizon over April 2007 to December 2025 and apply Romano–Wolf stepdown family-wise error rate control across the 42 asset $\times$ component cells. We find: (i) the three component innovations are statistically separable ($\rho_{\max} = 0.41$, $\text{VIF}_{\max} = 1.20$); (ii) 26 of 42 pre-specified asset–component cells reject the zero-coefficient null at RW-adjusted $p \leq 0.10$ with signs matching the locked sign map; and (iii) the absolute magnitude of each instrument’s loading on each component aligns with its locked duration / TIPS-hedge / curve-geometry prior at Spearman $\rho > 0.92$ in every component. The decomposition provides a public-data benchmark for fixed-income risk attribution that practitioners can rebuild daily without reliance on proprietary options-implied series.

JEL classification. G12, G13, E43, E44.

Keywords. Fixed-income risk attribution; TIPS decomposition; D’Amico–Kim–Wei (DKW); Gürkaynak–Sack–Wright (GSW); Romano–Wolf stepdown; Treasury ETF; Treasury futures.

1 Introduction

Cross-sectional risk attribution for liquid fixed-income vehicles typically relies on either (i) duration/convexity decompositions internal to a portfolio system, or (ii) options-implied volatility surfaces such as the MOVE index, which require proprietary inputs. We provide a public-data alternative: a low-dimensional, three-axis decomposition of asset-level daily returns onto model-implied real-rate innovations, model-fitted inflation-compensation innovations, and a residualized nominal-curve shape factor. All inputs are observable from public Federal Reserve releases.

Our empirical contribution is a pre-specified exposure map, reproducible from public data, for 14 liquid fixed-income instruments (10 ETFs and 4 Treasury futures) with family-wise error control across 42 asset-component cells, and per-cell magnitude translation to basis-point shocks for portfolio-level interpretation. Existing work using DKW and GSW components has predominantly studied inflation expectations and term-premium dynamics from a macro perspective; we apply the same components to the cross-section of tradeable instruments and add a residualized GSW-based curve-geometry axis as the third dimension.

The remainder of the paper proceeds as follows. Section 2 describes the data sources and the construction of the three components. Section 3 states the pre-specified hypotheses and tests. Section 4 reports separability, the per-cell sign map with Romano–Wolf adjusted p -values, and the magnitude ordering against locked priors. Section 5 translates the standardized β -coefficients into per-10bp basis-point exposures. Section 6 provides a transparent portfolio-attribution example using three baskets. Section 7 reports six robustness rows spanning four perturbation classes. Section 8 discusses interpretation and limitations. Section 9 concludes.

2 Data and component construction

2.1 Sources

GSW nominal yields.

The Gürkaynak et al. (2007) daily zero-coupon yield curve, fields `SVENY02`, `SVENY05`, `SVENY10`, `SVENY30`. Source file `feds200628.csv`, hash recorded in the data manifest.

DKW TIPS decomposition.

The D’Amico et al. (2018) TIPS-yield decomposition, updated by Kim et al. (2019), providing model-implied components of nominal and TIPS yields. Source file `DKW_updates.csv`, hash recorded in the data manifest.

Asset prices.

Public Yahoo Finance daily adjusted-close series for 14 locked tickers: `SHY`, `IEF`, `TLT`, `TIP`, `LQD`, `HYG`, `MBB`, `AGG`, `EMB`, `BIL` (10 ETFs), plus `ZT=F`, `ZF=F`, `ZN=F`, `ZB=F` (4 Treasury futures continuous front-month).

2.2 Component construction

We construct the three component series as follows.

Real-rate component R : model-implied true real yield equal to

$$R_t^{\text{level}} = w_5 \cdot (\text{exp.real.short.rate.5}_t + \text{real.term.prem.5}_t) + w_{10} \cdot (\text{exp.real.short.rate.10}_t + \text{real.term.prem.10}_t),$$

with common DV01-derived weights $w_5 = 0.232$ and $w_{10} = 0.768$ computed from the 14-asset duration table. The daily-change innovation in basis points is

$$\Delta R_t = 100 \cdot (R_t^{\text{level}} - R_{t-1}^{\text{level}}).$$

Inflation-compensation component (BEI): DKW model-fitted inflation compensation,

$$\text{BEI}_t^{\text{level}} = w_5 \cdot \text{ic.fitted.5}_t + w_{10} \cdot \text{ic.fitted.10}_t,$$

with the same weights, and ΔBEI_t defined analogously in basis points.

Shape component: a residualized nominal-curve geometry factor. Let ΔY_t^τ denote the daily change in the τ -year GSW nominal yield in basis points. Define

$$\text{Long}_t = \frac{1}{2} \Delta Y_t^{10} + \frac{1}{2} \Delta Y_t^{30}, \quad \text{Front}_t = \frac{1}{2} \Delta Y_t^{02} + \frac{1}{2} \Delta Y_t^{05},$$

and let $\text{Shape}_t^{\text{direct}} = \text{Long}_t - \text{Front}_t$. The orthogonalized Shape series Shape_t is the residual from

$$\text{Shape}_t^{\text{direct}} = a + b_R \Delta R_t + b_{\text{BEI}} \Delta \text{BEI}_t + u_t,$$

sign-normalized so that $\text{corr}(\text{Shape}_t, \text{Shape}_t^{\text{direct}}) > 0$.

All three component series are then z -scored over the primary sample (April 2007 to December 2025; $T = 4,481$ trading days). Per-day standard deviations of the raw bp innovations are $\sigma_R = 3.88$ bp, $\sigma_{\text{BEI}} = 3.19$ bp, and $\sigma_{\text{Shape}} = 3.76$ bp.

2.3 Asset returns

Daily log returns are

$$r_{i,t}^{(1)} = 100 \cdot (\log P_{i,t} - \log P_{i,t-1}),$$

in percent, using adjusted-close prices. The primary horizon is $h = 1$.

3 Hypotheses and pre-specified tests

H1 (separability). The three component innovations are statistically separable: $\rho_{\max} \leq 0.50$ AND $\text{VIF}_{\max} \leq 5$; the criterion is satisfied if both thresholds hold.

H2 (sign map). For each asset i and component k , we estimate the per-asset HAC Newey–West linear projection

$$r_{i,t}^{(1)} = \alpha_i + \beta_i^R R_t^z + \beta_i^{\text{BEI}} \text{BEI}_t^z + \beta_i^{\text{Shape}} \text{Shape}_t^z + \varepsilon_{i,t},$$

with HAC SE based on a Bartlett kernel and explicit lag $L_{\text{NW}}(T, 1) = \lfloor 4 \cdot (T/100)^{2/9} \rfloor = 9$. The 42 asset–component cell family is then subjected to Romano–Wolf stepdown family-wise error rate

control with $B = 5,000$ stationary block bootstrap resamples (block length $\ell = \lceil T^{1/3} \rceil = 17$, restart probability $p = 1/\ell$, seed 110503). H2 supports the primary path verdict if at least 60% of cells reject at RW-adjusted $p \leq 0.10$ with sign matching the locked sign map.

H3 (magnitude ordering). For each component, we compute the Spearman rank correlation between the absolute observed coefficient $|\beta_i|$ and a locked magnitude prior. The prior for R is each instrument’s effective duration (from issuer fact sheets and CME contract specifications); for BEI, the locked TIPS-hedge / credit-inflation prior; for Shape, the locked tenor-differential prior. H3 PASS for a component if Spearman $\rho > 0$ with bootstrap 95% CI strictly positive.

4 Results

4.1 H1 separability

Pairwise correlations of $(R^z, BEI^z, Shape^z)$ on the primary sample give $\max |\text{corr}| = 0.408$ for the (R, BEI) pair and approximately zero for the other two pairs (Shape is residualized by construction). Variance inflation factors are 1.20 for R , 1.20 for BEI, and 1.00 for Shape. The threshold $\rho_{\max} \leq 0.50$ AND $VIF_{\max} \leq 5$ holds: **H1: criterion satisfied**. The three components are statistically separable; R and BEI are not orthogonal but jointly identifiable. Figure 1 provides the pairwise scatter plots.

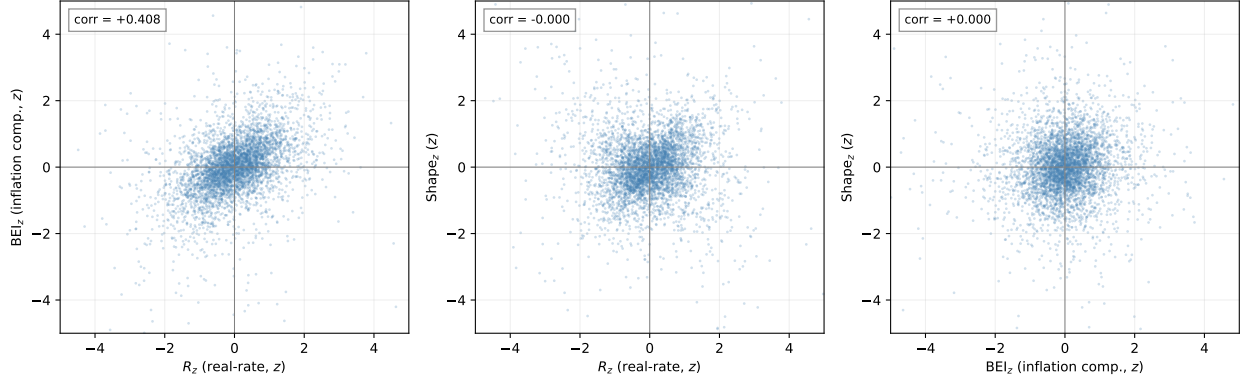


Figure 1: Pairwise scatter of the three standardized component innovations $(R^z, BEI^z, Shape^z)$ over the primary sample. The maximum absolute pairwise correlation is 0.408 (between R and BEI); Shape is residualized so its correlation with R and BEI is zero by construction.

4.2 H2 cell-level sign map

Table 1 reports the ten cells with largest $|t|$. All 14 asset R -component cells have negative coefficients with magnitudes broadly rank-aligned with effective duration. Across the full 42-cell family, 26 cells reject the zero-coefficient null at Romano–Wolf adjusted $p_{RW} \leq 0.10$, all with signs matching the locked sign map. This 26/42 ratio (61.9%) clears the pre-specified primary-path threshold of 60%.

Table 1: Top ten asset–component cells by $|t|$ in the primary 42-cell family. β in %-return per 1σ component shock; HAC Newey–West t ; Romano–Wolf adjusted p .

Rank	Asset	Component	β (z -units)	t	p_{RW}
1	ZN=F	R	−0.352	−107.7	< 0.001
2	IEF	R	−0.422	−98.9	< 0.001
3	ZF=F	R	−0.221	−95.2	< 0.001
4	ZB=F	R	−0.587	−81.3	< 0.001
5	TIP	R	−0.392	−70.8	< 0.001
6	SHY	R	−0.075	−45.5	< 0.001
7	ZT=F	R	−0.071	−44.5	< 0.001
8	TLT	R	−0.796	−43.7	< 0.001
9	TIP	BEI	+0.225	+34.4	< 0.001
10	TLT	Shape	−0.337	−32.1	< 0.001

The BEI point estimates for TIP and the credit-tilted ETFs (HYG, EMB, LQD) match the locked positive-BEI sign prior; formal cell-level claims are limited to RW-significant cells (TIP BEI is RW-significant, while HYG/EMB/LQD BEI cells are not). Long-end Treasuries (TLT, ZB) load negatively on Shape and the front-end (SHY, ZT, ZF) positively, both directions matching the locked Shape sign prior. BIL behaves as the locked near-null negative-control asset, with all three component estimates close to zero (R : $\beta = -0.0016$, $|t| = 1.7$; BEI: $\beta = -0.0015$, $|t| = 1.5$; Shape: $\beta = +0.0015$, $|t| = 1.4$).

Figure 2 maps the full 42-cell family of standardized exposures. The R column is uniformly negative and ordered by effective duration, the BEI column is positive for the inflation- and credit-sensitive instruments (TIP, EMB, HYG), and the Shape column separates long-end from front-end instruments.

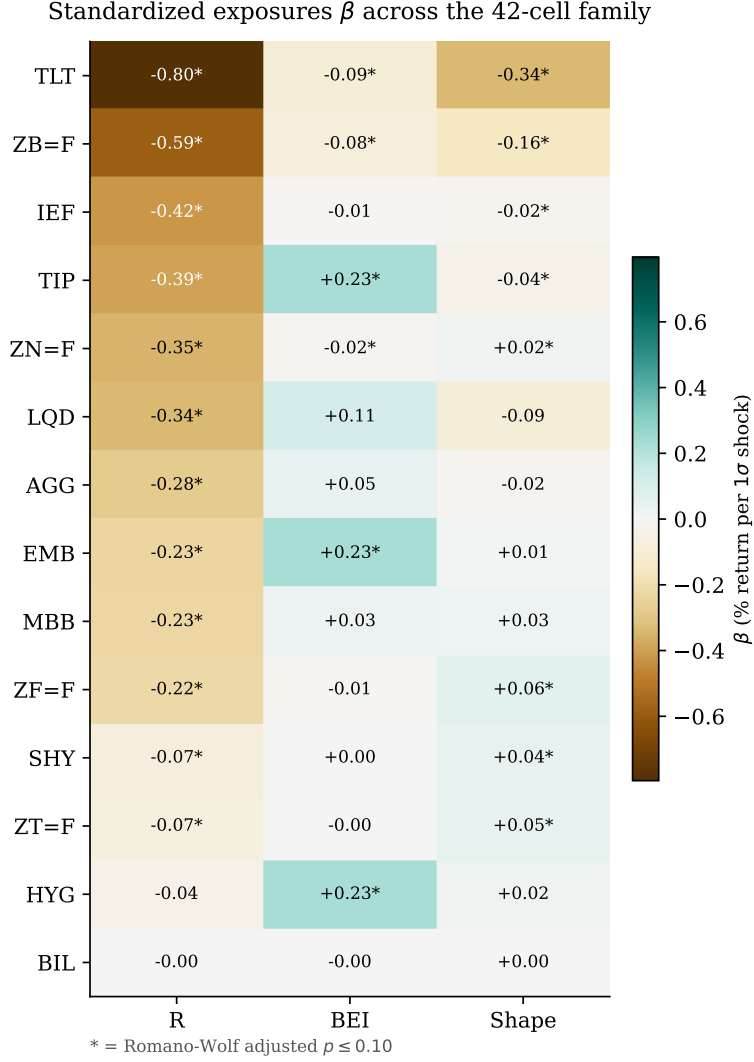


Figure 2: Standardized exposures β for the 42-cell family (14 assets $\times$ 3 components), in % return per 1σ component shock. Assets are ordered by the magnitude of their real-rate loading. An asterisk marks cells significant at Romano–Wolf adjusted $p \leq 0.10$.

4.3 H3 magnitude ordering

Table 2 reports the per-component Spearman rank correlation of $|\beta_{i,k}|$ against the locked magnitude prior, with bootstrap 95% confidence intervals (2,000 resamples).

All three component magnitude orderings are statistically distinguishable from zero at the 95% bootstrap level. Figure 3 provides the corresponding scatter plots.

Table 2: H3 magnitude ordering. Spearman ρ between $|\hat{\beta}_{i,k}|$ and locked magnitude prior; 95% bootstrap CI; verdict at $\alpha = 0.05$.

Component	Spearman ρ	95% bootstrap CI	Verdict
R	+0.921	[+0.695, +0.987]	PASS
BEI	+0.945	[+0.808, +0.985]	PASS
Shape	+0.932	[+0.730, +0.987]	PASS

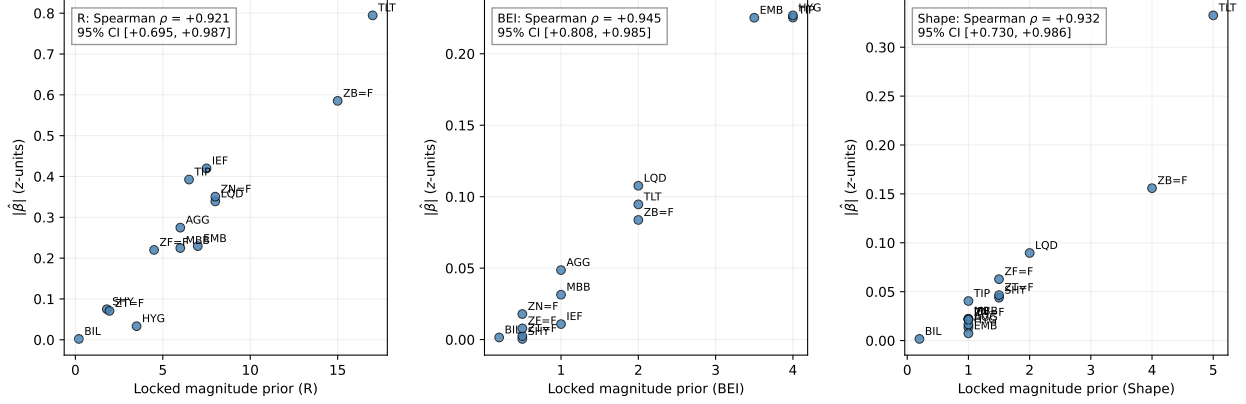


Figure 3: H3 magnitude ordering: observed $|\hat{\beta}_{i,k}|$ versus the locked magnitude prior for each component. Each point is one asset; labels identify the asset. Spearman ρ and 95% bootstrap confidence intervals reported in each panel title.

5 Magnitude translation: z -score β to basis points

Each β in Table 1 is reported in z -score units, i.e., % return per 1σ component shock. A practitioner-anchored translation per 10 bp shock to a single component is given by

$$\frac{\Delta r_{i,k}}{\Delta k_{\text{bp}} = 10 \text{ bp}} = \beta_{i,k}^z \cdot \frac{10}{\sigma_k^{\text{bp/d}}},$$

in % daily return per 10 bp shock to component k . Table 3 reports this translation for all 14 assets and all three components.

The R column tracks duration-like exposure with the expected sign: TLT's -2.05% per +10 bp R shock matches a duration of approximately 20 years; SHY's -0.19% matches a duration of approximately 1.8 years. The BEI coefficient captures a reduced-form credit/inflation co-movement in the primary sample: TIP, HYG, EMB, and LQD all have positive BEI exposures, with TIP and HYG/EMB the largest. The Shape column captures long-end relative repricing (TLT -0.90% , ZB -0.42%) versus front-end repricing (SHY $+0.12\%$, ZT $+0.12\%$). Figure 4 visualizes the per-asset basis-point translation across the three component shocks.

Table 3: Magnitude translation: % daily return per +10 bp shock to a single component (point estimates from primary sample). Other components held fixed.

Asset	per +10 bp R	per +10 bp BEI	per +10 bp Shape
TLT	−2.05%	−0.30%	−0.90%
ZB=F	−1.51%	−0.26%	−0.42%
IEF	−1.09%	−0.03%	−0.06%
TIP	−1.01%	+0.71%	−0.11%
ZN=F	−0.91%	−0.05%	+0.06%
LQD	−0.88%	+0.34%	−0.24%
AGG	−0.72%	+0.16%	−0.04%
EMB	−0.60%	+0.71%	+0.02%
MBB	−0.59%	+0.10%	+0.07%
ZF=F	−0.57%	−0.02%	+0.17%
SHY	−0.19%	+0.00%	+0.12%
ZT=F	−0.18%	−0.01%	+0.12%
HYG	−0.10%	+0.72%	+0.05%
BIL	−0.004%	−0.005%	+0.004%

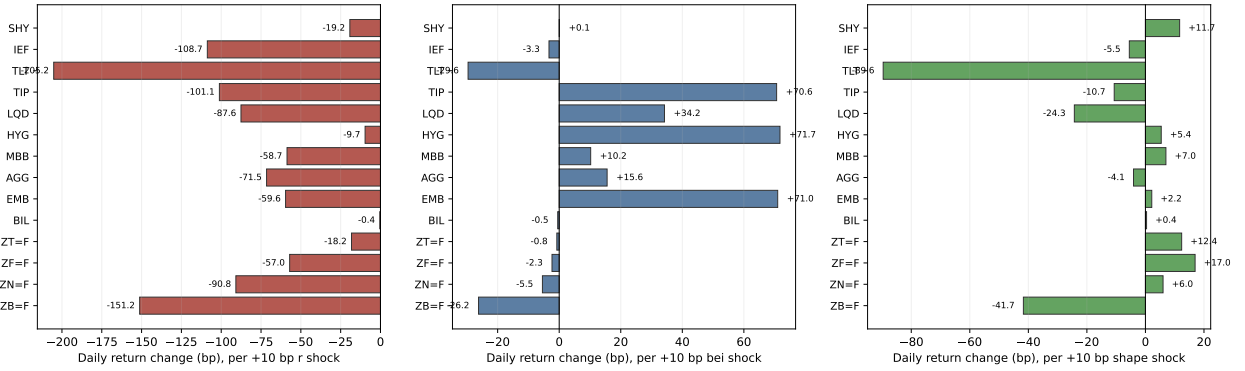


Figure 4: Magnitude translation: daily return change in basis points (1% = 100 bp) per +10 bp shock to a single component, holding the other two components fixed. Point estimates from the primary sample.

6 Portfolio attribution example

To anchor the practitioner reading without optimizing or backtesting, we apply the per-cell β estimates to three transparent fixed-income baskets and decompose fitted daily-return contributions into R , BEI, and Shape components. We note that the R –BEI covariance term is non-zero because R and BEI have non-zero pairwise correlation (0.41), so basket-level variance shares cannot be cleanly partitioned by component.

Baskets.

- *Bond Sleeve* (canonical 60/40 fixed-income tilt): SHY 30%, IEF 30%, AGG 25%, TIP 15%.
- *Duration Heavy*: TLT 50%, ZB=F 20%, IEF 20%, ZN=F 10%.
- *Credit Tilted*: HYG 30%, EMB 25%, LQD 25%, AGG 20%.

Table 4: Basket-level component β from weighted average of per-cell estimates.

Basket	β_R	β_{BEI}	β_{Shape}
Bond Sleeve	−0.276	+0.043	−0.004
Duration Heavy	−0.633	−0.068	−0.200
Credit Tilted	−0.207	+0.161	−0.020

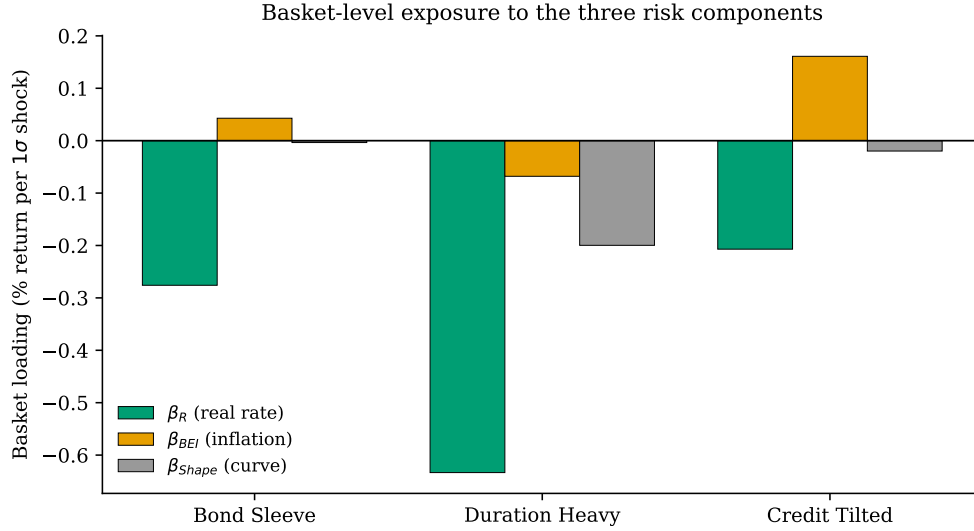


Figure 5: Basket-level loadings on the three risk components, in % return per 1σ shock, from the weighted average of per-cell estimates. The real-rate component dominates the Duration Heavy basket, while the inflation component is the largest positive loading for the Credit Tilted basket.

Basket-level β (% return per 1σ shock).

Realized-day attribution. The illustrative day is selected mechanically as the largest absolute Bond Sleeve daily return in calendar 2024, which is 2 August 2024; the selection rule was fixed before inspecting the decomposition, and the example is not used for inference or model validation. On that day the component innovations were $R^z = -3.57$, $BEI^z = -4.90$, $Shape^z = +3.57$. Fitted attributions:

The R channel dominates the Treasury-heavy baskets; the BEI coefficient is the largest-magnitude single fitted contributor for the credit-tilted basket. For the Duration Heavy basket on this day, Shape offsets the R /BEI fitted contribution by -0.71% , about 30% of the actual return in absolute

Table 5: Realized-day attribution for 2 August 2024. Attribution columns are basket-level fitted contributions; total is the sum; residual is actual minus fitted.

Basket	Actual ret.	Attrib R	Attrib BEI	Attrib Shape	Attrib total	Residual
Bond Sleeve	+0.98%	+0.98%	−0.21%	−0.01%	+0.76%	+0.22%
Duration Heavy	+2.37%	+2.26%	+0.33%	−0.71%	+1.88%	+0.50%
Credit Tilted	+0.53%	+0.74%	−0.79%	−0.07%	−0.12%	+0.65%

value. The Credit Tilted residual is large that day because HYG outperformed its R -implied trajectory, consistent with credit-spread or risk-sentiment moves not captured by the triangle.

This example is a transparent point-in-time attribution, not an optimized strategy or backtest.

7 Robustness

We pre-specified four robustness perturbation classes, yielding six specification rows ($h = 5$, $h = 20$, AR(1), pre-2010, post-2010, and PC1-Shape), to test sign stability of the 42-cell family. Across the 30 cells achieving $|t| > 3$ in the primary specification, sign agreement averages 97% across the six rows (Table 6). The robustness package supports sign stability of the primary architecture; it does not re-estimate Romano–Wolf at each perturbation.

Table 6: Robustness: sign-preservation against the 30 primary-strong cells ($|t| > 3$ in primary). All-cell sign match also reported.

Robustness perturbation	Strong-cell sign match	All-cell sign match
$h = 5$ multi-horizon	28/30 (93%)	36/42
$h = 20$ multi-horizon	28/30 (93%)	36/42
AR(1) residual innovations	30/30 (100%)	41/42
Pre-2010 sub-sample ($T \approx 653$)	30/30 (100%)	39/42
Post-2010 sub-sample ($T \approx 3,822$)	30/30 (100%)	42/42
Direct slope/curv PC1 Shape	29/30 (97%)	39/42

The aggregate strong-cell sign preservation across the six robustness rows is $175/180 = 97.2\%$. The few discordant cells are uniformly near-null in the primary specification ($|\hat{\beta}_{\text{prim}}| < 0.025$), so the robustness mismatches occur within the noise floor of the original signal rather than in the primary signal itself. Figure 6 visualizes the sign-preservation pattern across the 30 strong cells and the six robustness rows.

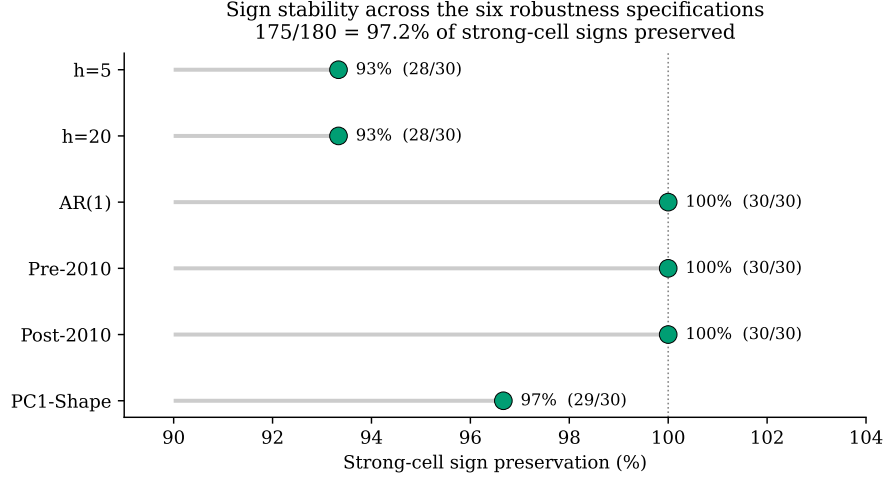


Figure 6: Sign preservation of the 30 primary-strong cells ($|t| > 3$) across the six robustness rows. Blue marks a cell whose sign matches the primary specification; red marks a sign flip. Of 180 cell-by-row comparisons, 175 (97.2%) preserve sign, and every flip occurs in a cell that is near-null in the primary specification.

Multi-horizon $h = 5$ and $h = 20$. HAC lag set to $L_{NW}(T, h) = \max(h - 1, \lfloor 4(T/100)^{2/9} \rfloor)$ to absorb the overlap induced by the cumulative-return construction.

AR(1) innovations. We extract AR(1) residuals from the raw component changes ($\hat{\phi}_R = +0.008$, $\hat{\phi}_{BEI} = +0.233$, $\hat{\phi}_{Shape} = -0.020$) and re-estimate the per-asset HAC LP. The top-10 cells by $|t|$ are unchanged in sign and magnitude.

Pre/post-2010 sub-sample. The pre-2010 sub-sample contains the global financial crisis and early TIPS-market thinness; the post-2010 sub-sample covers the QE era and post-COVID. Both preserve all 30 strong cells' signs.

Direct slope/curv PC1 Shape. The first principal component of $(\Delta\text{slope}, \Delta\text{curv})$ from the GSW nominal curve explains 54.7% of slope+curv variance; loadings are $(+0.707, +0.707)$. 29/30 strong-cell sign match (the lone flip is the ZN=F Shape cell, with $|\hat{\beta}_{\text{prim}}| = 0.021$, at the noise floor).

8 Discussion and limitations

The triangle (R , BEI, Shape) reduces the daily co-movement of 14 liquid fixed-income instruments to three publicly observable, low-collinearity exposures. The empirical pattern is largely a confirmation of duration, TIPS-hedge, and curve-geometry priors, but those priors are now jointly tested with

family-wise error control on a single pre-specified design, with quantitative magnitude translation directly to basis-point shocks.

For practitioners, the triangle is most directly useful as (i) an attribution lens (Section 6); (ii) a magnitude anchor for portfolio-level shock scenarios (Section 5); and (iii) a public-data exposure-mapping benchmark that does not use options-implied volatility inputs. The decomposition does not itself forecast or predict; it is a contemporaneous mapping of return co-movement to component innovations.

For academics, the triangle is a parsimonious low-dimensional decomposition of the cross-section that complements the macro literature on TIPS decomposition by mapping the same components onto the cross-section of tradeable instruments. Open extensions include direct MOVE / realized-volatility validation, FOMC and CPI event-window narrative-identification, and full propagation of generated-regressor uncertainty for the DKW components.

Limitations.

- (1) R and BEI are model-generated regressors from DKW; we treat them as observed. Generated-regressor uncertainty is not propagated into the HAC standard errors. This is a known limitation of using staff-model components on the right-hand side.
- (2) Pairwise $\text{corr}(R, \text{BEI}) = 0.41$, below the H1 separability threshold but not zero. R and BEI are jointly identifiable but not orthogonal; coefficients are partial conditional on the other.
- (3) Robustness is sign stability, not full family-wise re-testing at each perturbation. We do not re-estimate Romano–Wolf at $h = 5$, $h = 20$, $\text{AR}(1)$, etc.; we report sign preservation against the primary 42-cell decision set.
- (4) H2 clears the pre-specified 60% threshold by a narrow margin ($26/42 = 61.9\%$).
- (5) Yahoo Finance adjusted-close prices and continuous front-month futures are public and reproducible but not institution-grade (no roll management for futures; pre-2012 dividend reinvestment for some ETFs may be imperfect).
- (6) Multi-horizon $h = 5$ and $h = 20$ returns overlap by construction, partially absorbed by the HAC lag $L_{\text{NW}}(T, h) = \max(h - 1, \lfloor 4(T/100)^{2/9} \rfloor)$. Horizon-specific family-wise re-testing would be a useful extension.
- (7) No MOVE / realized-volatility validation is performed. The paper makes no claim that the triangle reproduces the options-implied volatility structure; the framing is exposure mapping, not volatility decomposition.

9 Conclusion

The Treasury Risk Triangle reduces 14 liquid fixed-income instruments' daily co-movement to three low-collinearity, publicly observable risk components: model-implied real-yield change, model-fitted inflation compensation change, and residualized nominal-curve geometry. Sign and magnitude evidence at the 5,000-bootstrap Romano–Wolf level supports the primary criterion: 26 of 42 pre-specified cells survive family-wise error control, all three components pass magnitude ordering at Spearman $\rho > 0.92$, and 175 of 180 strong-cell signs are preserved across six rows spanning four pre-specified perturbation classes.

References

- Stefania D'Amico, Don H. Kim, and Min Wei. Tips from TIPS: The informational content of Treasury inflation-protected security prices. *Journal of Financial and Quantitative Analysis*, 53(1):395–436, 2018. doi: 10.1017/S0022109017001077.
- Refet S. Gürkaynak, Brian Sack, and Jonathan H. Wright. The U.S. Treasury yield curve: 1961 to the present. *Journal of Monetary Economics*, 54(8):2291–2304, 2007. doi: 10.1016/j.jmoneco.2007.06.029.
- Don H. Kim, Cait Walsh, and Min Wei. Tips from TIPS: update. FEDS Notes, Federal Reserve Board, 2019.